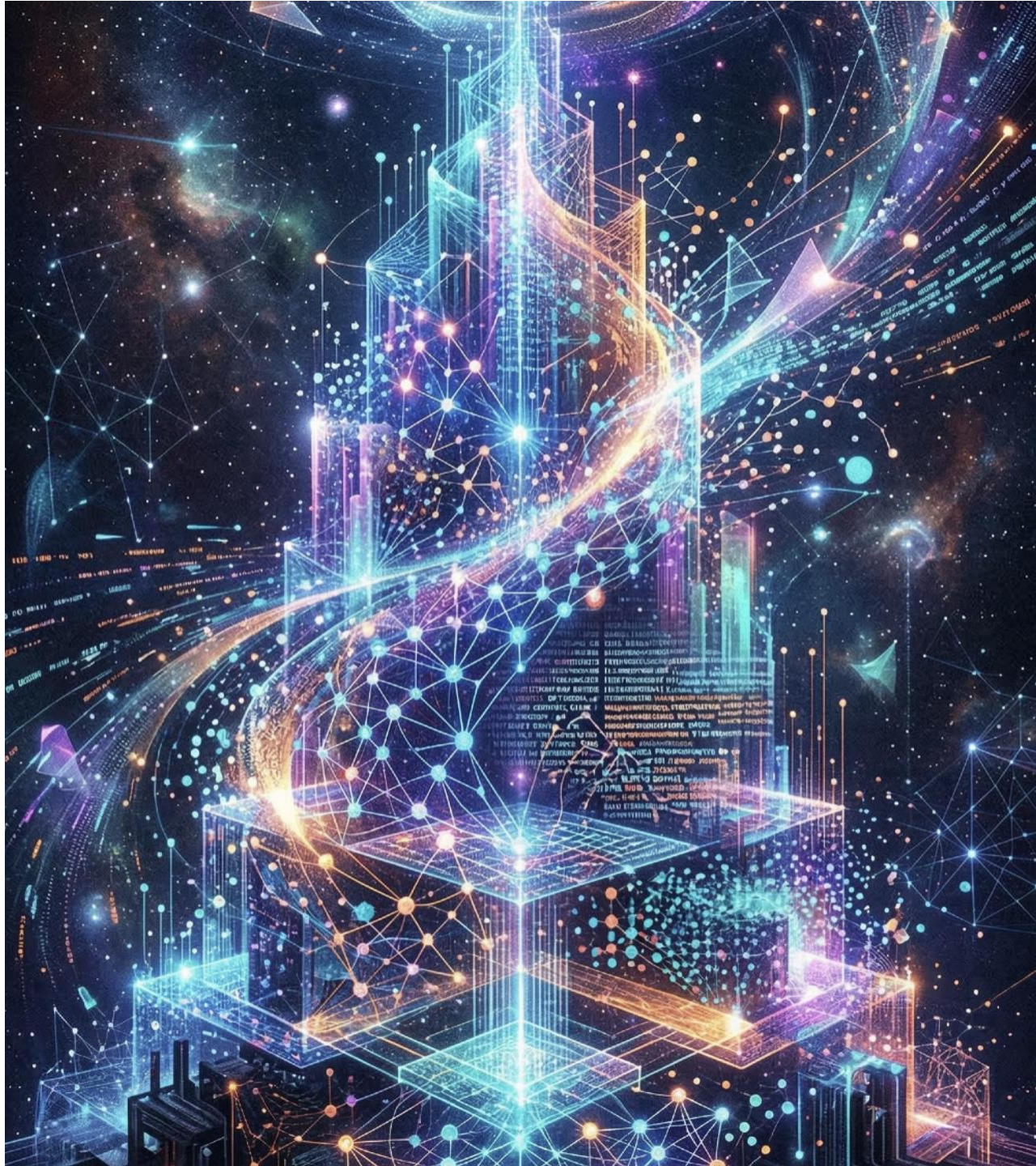


Hochschule  
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# Large Language Models & Foundation Models

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# Kann ein selbstgebautes LLM sinnvolle Kochrezept generieren?

- **Datensätze** – Auswahl & Aufbereitung
- **Pre-Training** – Training & Ergebnisse
- **Fine Tuning** – Klassifikation & Instruction
- **RAG** – Retrieval & Generation

# Eigenes Modell

# RecipeNLG Datensatz

## Unser Pre-Training Korpus

- **2,23 Mio. Rezepte**

Veröffentlichung (2020) größter öffentliche Rezept-Datensatz

- **Semi-strukturierter Aufbau**

| Title                      | Ingredients                                                                                                 | Directions                                                      | Link                                                                                                                 | Source   | NER                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------|
| Kids' Strawberry Margarita | ["4 or 5 large fresh or frozen strawberries", "1 c. lemonade or limeade", "squeeze of lime", "crushed ice"] | ["Blend all ingredients at high speed and serve with a straw."] | <a href="http://www.cookbooks.com/Recipe-Details.aspx?id=887403">www.cookbooks.com/Recipe-Details.aspx?id=887403</a> | Gathered | ["frozen strawberries", "limeade", "lime", "crushed ice"] |

# RecipeNLG Datensatz

Warum dieser Datensatz?

- **Richtige Domäne**
- **(Semi-)strukturiert**
- **Aber inkl. Freitexte**

| Title                      | Ingrediens                                                                                                  | Directions                                                      |
|----------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Kids' Strawberry Margarita | ["4 or 5 large fresh or frozen strawberries", "1 c. lemonade or limeade", "squeeze of lime", "crushed ice"] | ["Blend all ingredients at high speed and serve with a straw."] |



# RecipeNLG Datensatz

## Vom CSV zum Trainingskorpus

- **Transformation**

pro Rezept:

Recipe: {title}

Ingredients:

{eine Zutat pro Zeile}

Directions:

{ein Schritt pro Zeile}

- **Trainingskorpus**

alle Rezepte verbunden,  
getrennt durch <|endoftext|>

## Trainingsformat

**Recipe:** Kids' Strawberry Margarita

**Ingredients:**

4 or 5 large fresh or frozen

1 c. lemonade or limeade

squeeze of lime

crushed ice

**Directions:**

Blend all ingredients at high speed and serve with a straw

<|endoftext|>

**Recipe:** Baked Beans

**Ingredients:**

3 (1 lb.) cans pork and beans

1/2 c. bell pepper, chopped

...

# Pre-Training

# Modellkonfiguration und DataLoader

## Modellkonfiguration

|                      |                         |
|----------------------|-------------------------|
| vocab_size = 50257   | → gpt2 Tokenizer        |
| context_length = 512 | → reduziert von 1024    |
| emb_dim = 768        | → Dimensionen pro Token |
| n_heads = 12         | → $768 \div 12 = 64$    |
| n_layers = 12        | → TransformerBlöcke     |
| drop_rate = 0.1      | → 10% Dropout           |

## DataLoader

|                  |                                    |
|------------------|------------------------------------|
| batch_size = 4   | → 4 Sequenzen pro Schritt          |
| max_length = 256 | → Fenstergröße                     |
| stride = 256     | → keine Überlappung                |
| shuffle = True   | → zufällige Reihenfolge            |
| drop_last = True | → unvollständige Batches verwerfen |
| num_epoche = 7   | → ein Durchlauf                    |

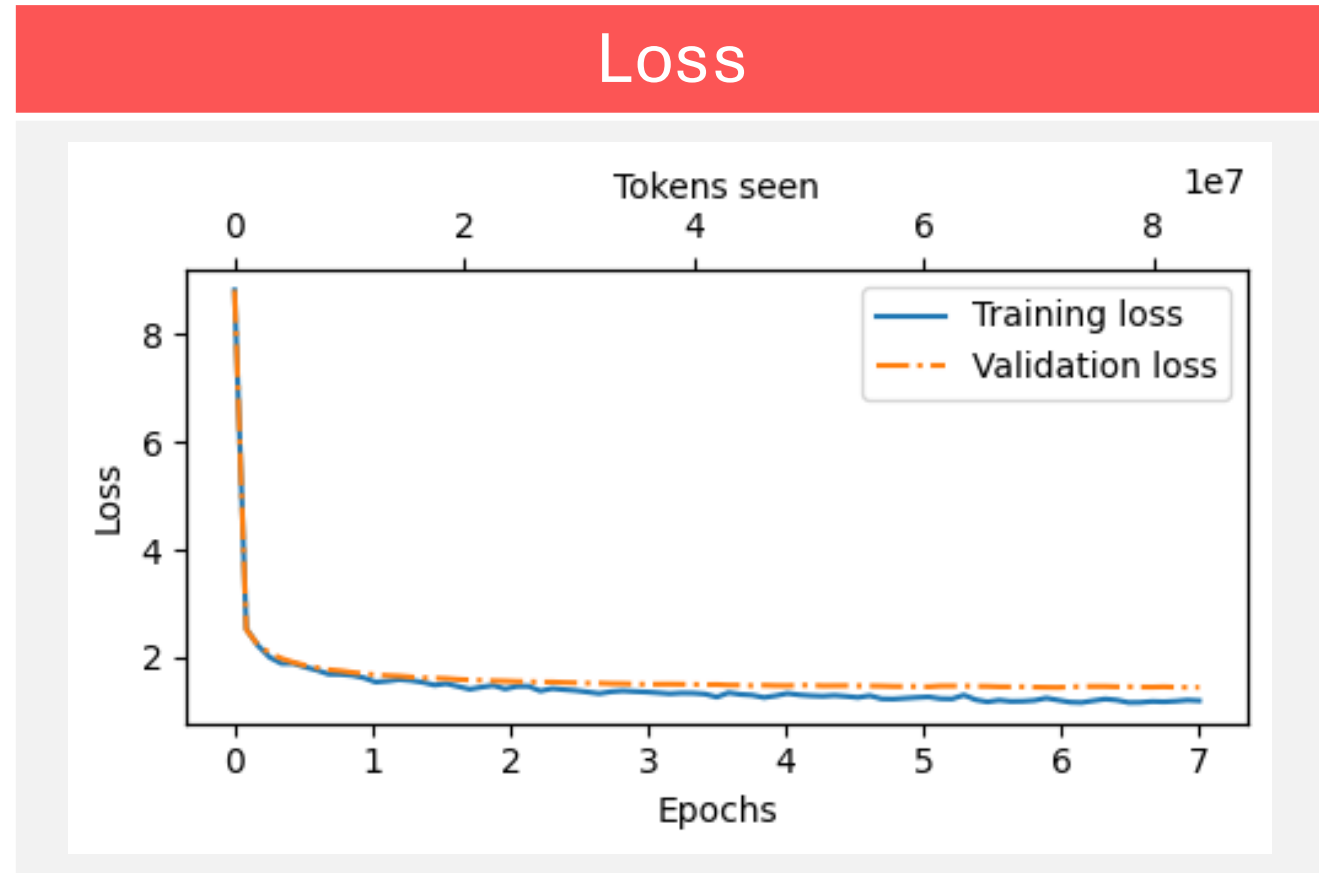


# Trainingsverlauf: Loss Kurve

Initialer Loss bei 10.89 → Niveau zufälliger Prediction

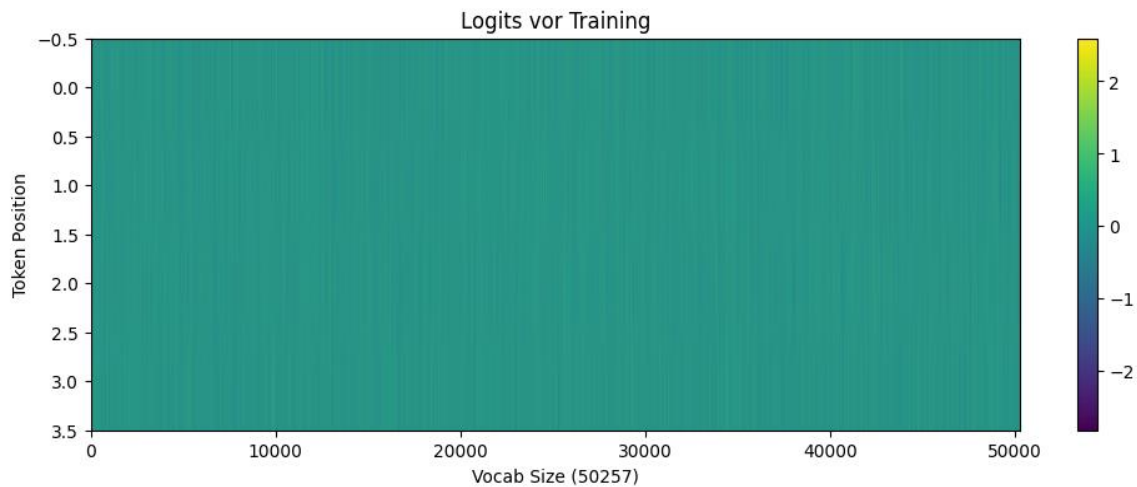
- Sehr steiler Verlustabfall zu Beginn
- Stabilisierung des Training Loss bei  $\sim 1.198$
- Validation Loss leicht erhöht bei  $\sim 1.447$
- Kleine, konstante Generalization Gap im Endbereich
- Keine Divergenz der Kurven → robustes Lernverhalten
- Reduktion : 78%
- Verbesserung: Faktor 4646x

Das bedeutet: Das Modell weißt dem richtigen Token nach dem Training eine 4646 mal höhere Wahrscheinlichkeit zu als vor dem Training

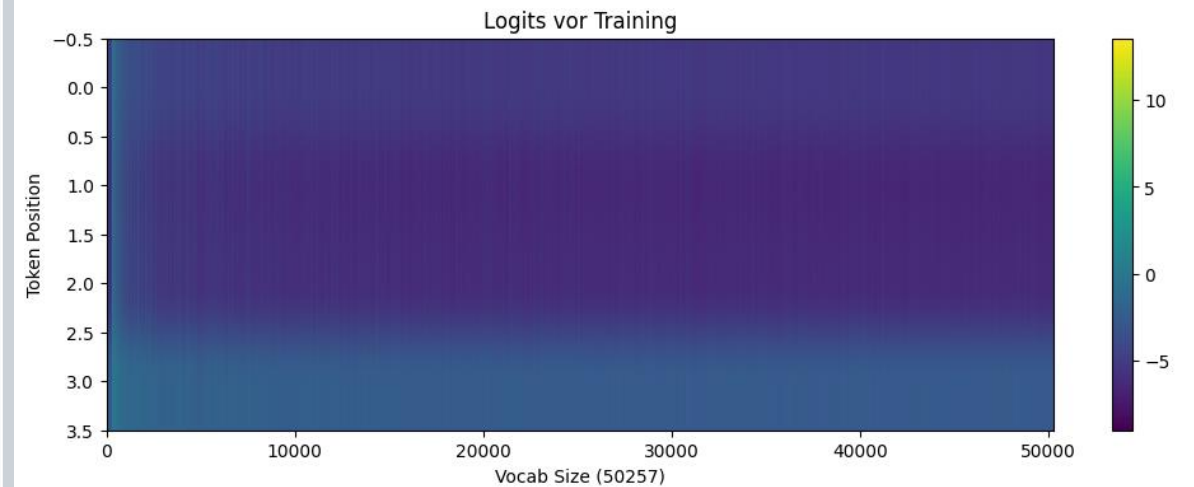


# Logits: Vorher/Nachher Vergleich

Logits vor dem Training

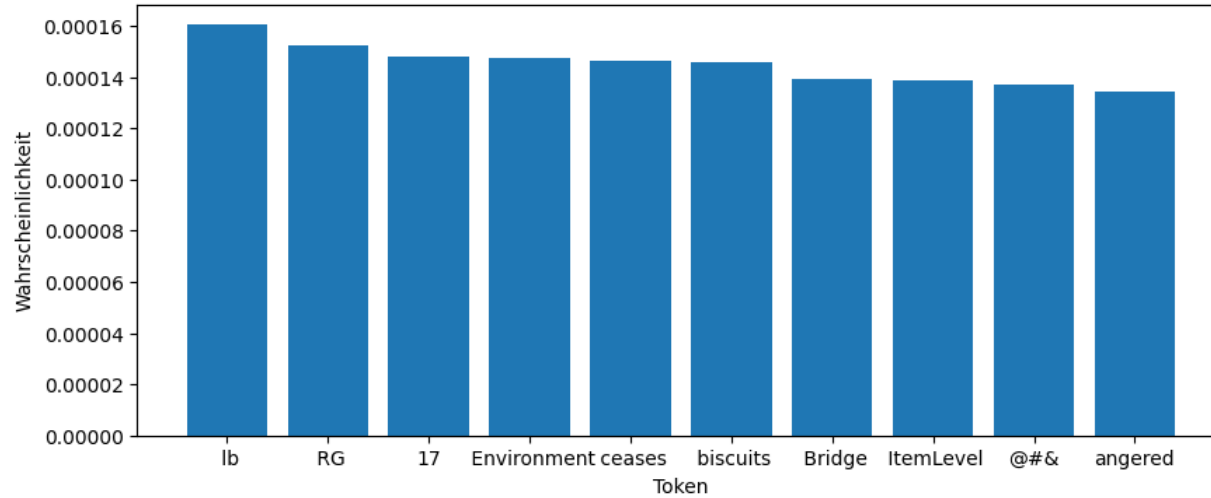


Logits nach dem Training

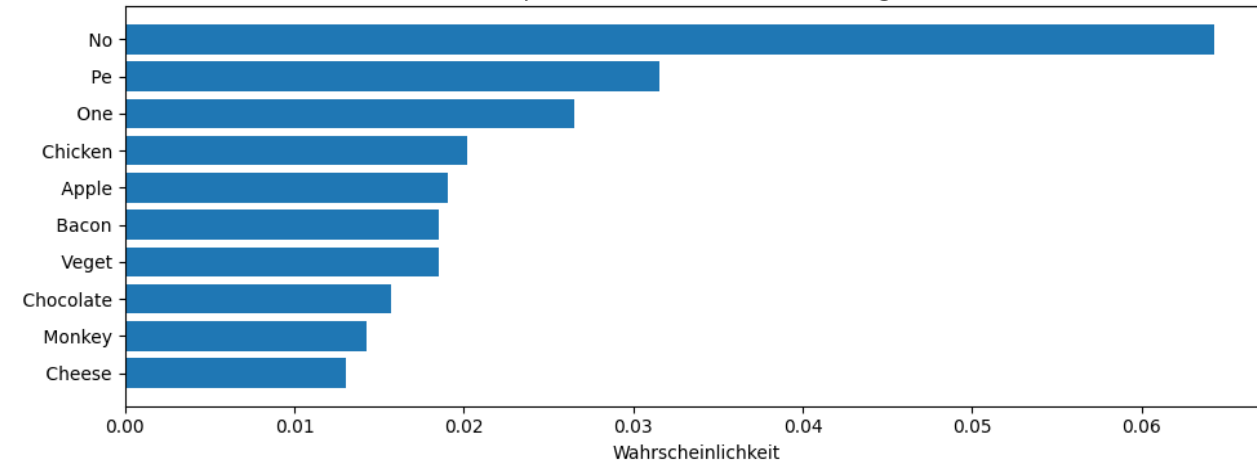


# Wahrscheinlichkeit des nächsten Tokens

Top 10 wahrscheinlichste nächste Tokens vor Training

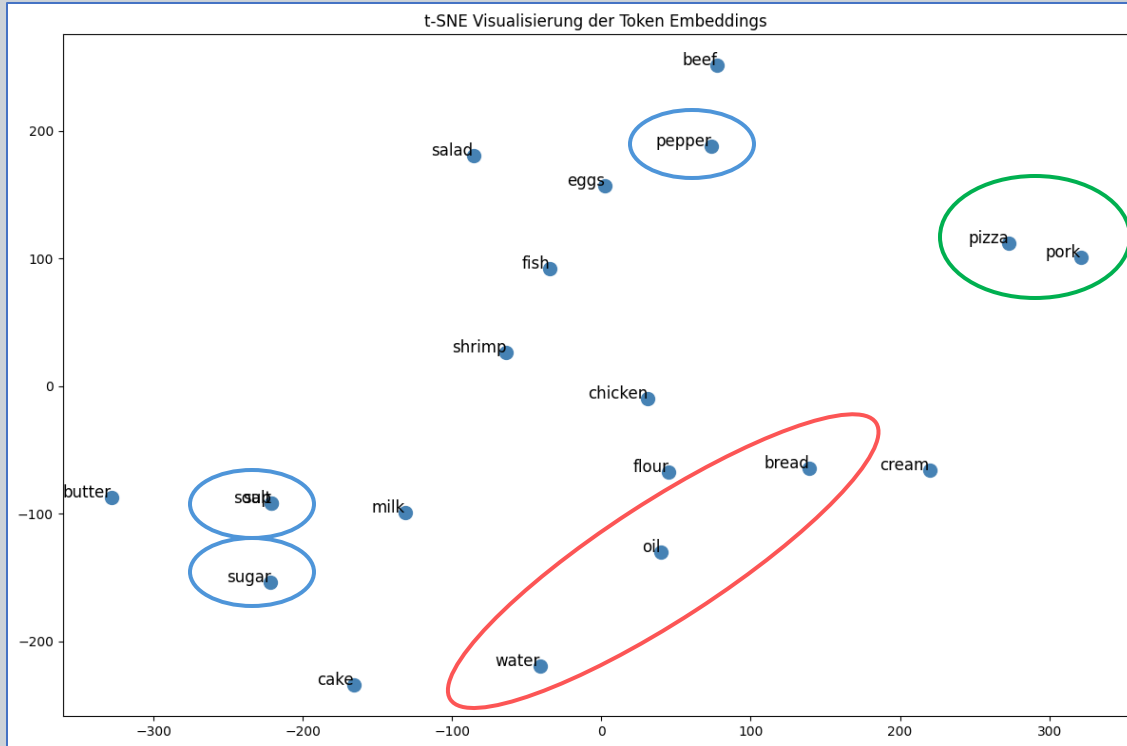


Top 10 nächste Tokens nach Training



# t-SNE: Visualisierung der Token Embeddings

## Vor dem Training



## Nach dem Training



# From-Scratch vs. OpenAI GPT-2

## Unser Modell – Text Output

Recepie: Chocolate Cake

Ingredients:

2 3/2 c. sugar

4 c. sugar

3 Tbsp. sugar

vanilla, cut into 1/4 c. sugar

8 oz. chocolate bits

4 c. flour

2 c. sliced or use out

unbaked pie shell

8 oz. whipped topping

2 c. cold

1/4 c. sliced peaches

4 tsp. lemon, softened

1/4 large can mandarin oranges

3/2 tsp. dried potatoes soup

## GPT-2 – Text Output

Recepie: Chocolate Cake

Ingredients:

For the filling:

Creamy almond milk, 2 tbsp. cold water, 1 tsp. vanilla extract

Directions:

For the buttercream icing:

Use 2 tbsp. of buttercream icing mix (I use 1 tbsp. of pure honey)

to form a thin layer of almond milk and whisk with the vanilla extract.

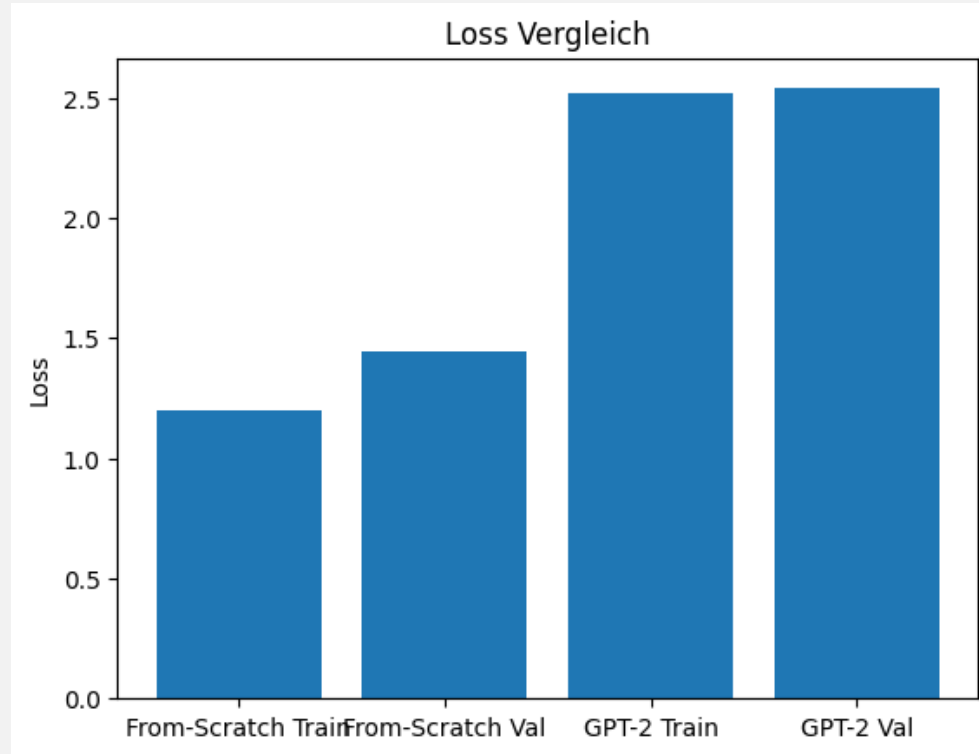
Spread evenly onto the butter cake.

For the icing cream:

Put 1/4 cup cold water into a bowl and fill

# From-Scratch vs. OpenAI GPT-2

## Loss Vergleich



# Context

| Choclate Cake                                                                                                                                                                                                                                                                                                                                                                                    | Pizza                                                                                                                                                                                                                                                                                                                                                             | Soup                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Recepie: Chocolate Cake</p> <p>Ingredients:</p> <p>7 3 egg yolk (28 oz.) can crushed tomatoes (6 oz. chopped green beans, drained, cut) Cool Whip dash olives, drained and drained 2 Tbsp. grated Cheddar cheese mix 4 c. sour cream 1/4 Tbsp. cornstarch pepper</p> <p>Directions: Cook in a 9 x 9-inch pan and bake about 1/2 of bread and bake at 350° in pan and 15 minutes (about 25</p> | <p>Recepie: Pizza</p> <p>Ingredients:</p> <p>7 3 egg yolk (28 oz. can crushed tomatoes (6 oz. chopped green beans, drained, cut) pepperoni, pressed potatoes cheese, drained 2 Tbsp. grated Cheddar cheese cubes-Up, drained 2 Tbsp. butter broth, cubed, dissolved (optional) Directions: Cook in a layer and mix until meat is well, stirring to 3/4 hours.</p> | <p>Recepie: Soup</p> <p>Ingredients:</p> <p>7 3 egg yolk (28 oz. can crushed tomatoes (6 oz. chopped green beans, drained, cut) pepperoni, pressed, drained fresh) 2 Tbsp. grated Cheddar cheese cubes-Up, drained 2 (11) c. sugar, whipped topping 2 can condensed. lemon juice 3 eggs 6 oz.) 4 Tbsp. cold water (20 drops onions, drained 6 hard-Tel tomatoes, rins (about 2</p> |



# Classification Fine Tuning

## 3A2M

# 3A2M Cooking Recipe Datensatz

## Der Klassifikations Datensatz

- **RecipeNLG als Basis**

- **Labeling**

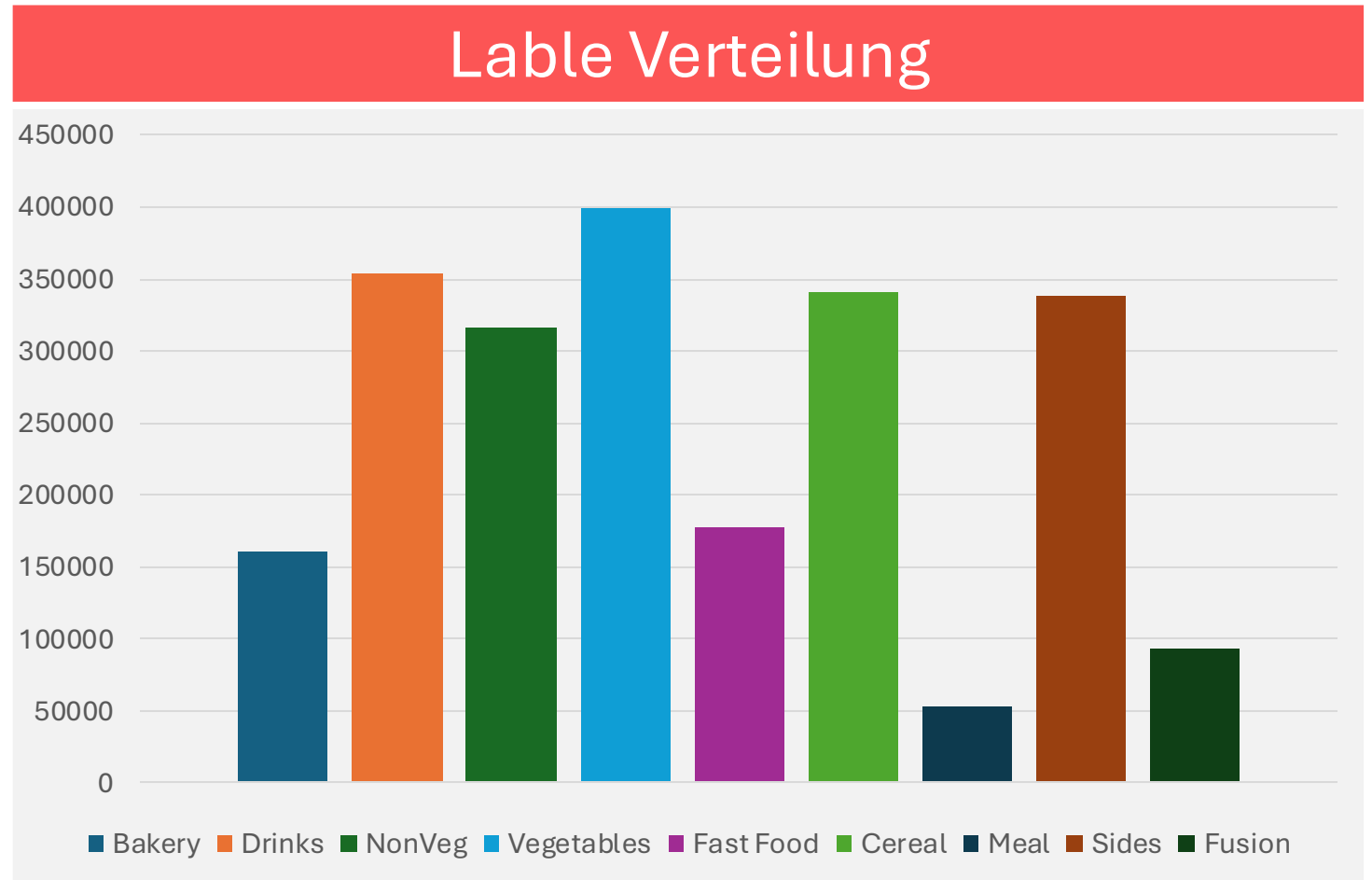
300k von Hand → restliche 1,9 Mio. per Active Learning

| Title                      | Directions                                                      | NER                                                       | Genre  | Lable |
|----------------------------|-----------------------------------------------------------------|-----------------------------------------------------------|--------|-------|
| Kids' Strawberry Margarita | ["Blend all ingredients at high speed and serve with a straw."] | ["frozen strawberries", "limeade", "lime", "crushed ice"] | drinks | 2     |

# 3A2M Cooking Recipe Datensatz

## Genres

1. Bakery
2. Drinks
3. NonVeg
4. Vegetables
5. Fast Food
6. Cereal
7. Meal
8. Sides
9. Fusion



# 3A2M Cooking Recipe Datensatz

## 1. String bauen

- full\_text

## 2. Rezepte filtern

- >1024 Tokens raus (GPT-2 Kontextlimit)

## 3. Ausgleich der Lableverteilung

- Undersampling

## 4. Split

- → 70 / 10 / 20 (Train/Val/Test)

### Klassifikations-Datensatz

| full_text                                                                                                                                                                                         | lable | genre  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------|
| <b>Recipe:</b> Kids' Strawberry Margarita<br><b>Ingredients:</b> frozen strawberries<br>limeade lime crushed ice<br><b>Directions:</b> Blend all ingredients at high speed and serve with a straw | 2     | drinks |
| <b>Recipe:</b> Baked Beans<br><b>Ingredients:</b> 3 (1 lb.) cans pork and beans 1/2 c. bell pepper, chopped ...                                                                                   | 3     | NonVeg |

# Initial Test

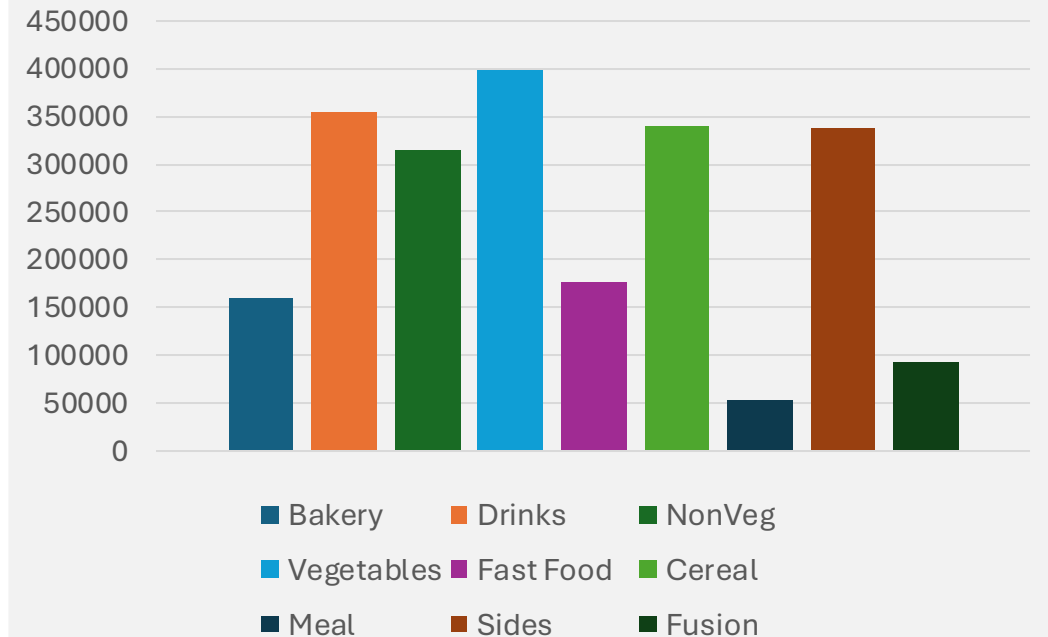
- Basis Modell: OpenAI-GPT2

| Test             | Input                                                                                                                                                                                                                                                                     | Output                                                                              | Kommentar                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Text Generierung | Recepie: Jewell Ball'S Chicken                                                                                                                                                                                                                                            | [00:00:00]EMOTE: *no key*/(                                                         | – keine sinnvolle Textgenerierung                                                                                                           |
| Klassifizierung  | <p>What would you label the following recipe? Answer with 'bakery', 'drinks', 'nonveg', 'vegetables', 'fastfood', 'cereal', 'meal', 'sides', or 'fusion':</p> <p>Recipe: Meyer Lemon Marmalade Ingredients: lemons, sugar Directions: Rinse the lemons and pat dry...</p> | The first time I saw the new "The Walking Dead" trailer, I was so excited. I was so | <ul style="list-style-type: none"><li>– Keine Klassifizierung</li><li>– Thema verfehlt</li><li>– keine sinnvolle Text fortsetzung</li></ul> |

# Multi- vs. Single-Label

- **Multi-Label Klassifikation**
  - Mehrere Labels pro Rezept
  - Nicht möglich mit dem Datensatz
- **Single-Label Klassifikation:**
  - Genau 1 Label pro Rezept
  - Genauigkeit vor dem Fine Tuning:
    - Training: 11,25%
    - Validation: 18,5%
    - Test: 17,5%
    - **Zufallsauswahl:** 11,11% (1/9)

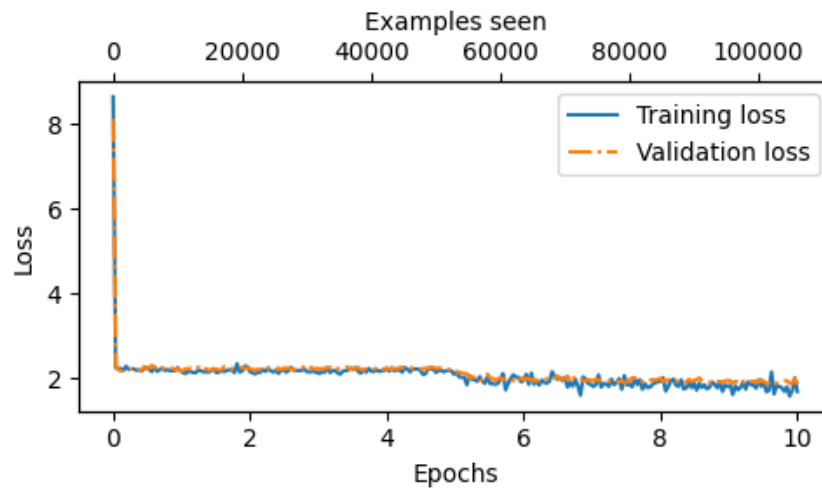
Lable Verteilung



# Training

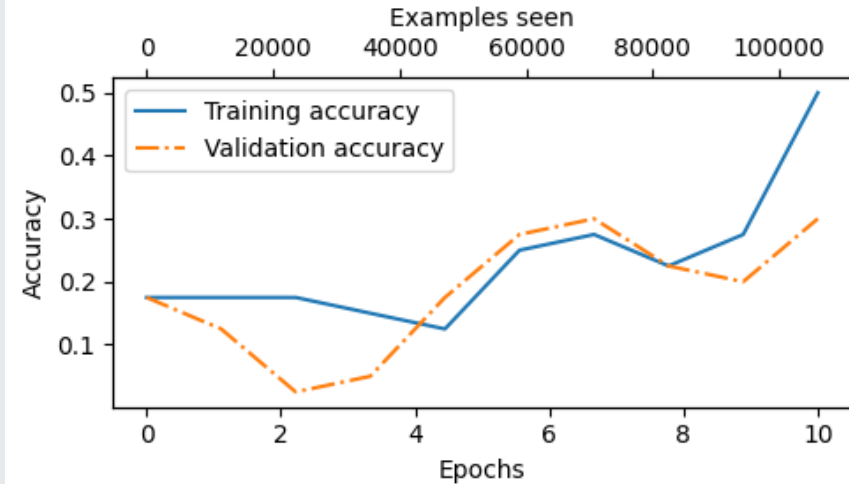
## Loss

- Start: 8,636
- Ende: 1,86



## Accuracy

- Start: 11,25%
- Ende: 32,87%



# Evaluierung

|                         |                                                                                                                                           |                                                                                                                                  |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Text Input</b>       | <b>Recipe:</b> Meyer Lemon Marmalade<br><b>Ingredients:</b> lemons, sugar<br><b>Directions:</b> Rinse the lemons and pat dry...           | Hey, just wanted to check if we're still on for dinner tonight? Let me know!                                                     |
| <b>Output</b>           | <b>fusion</b>                                                                                                                             | <b>fusion</b>                                                                                                                    |
| <b>Richtig / Falsch</b> | <b>Falsch</b>                                                                                                                             | <b>Falsch</b>                                                                                                                    |
| <b>Mögliche Gründe</b>  | <ul style="list-style-type: none"><li>• Bereits unsauberes Labeling im Datensatz → Marmalade ist als <b>Drink</b> Klassifiziert</li></ul> | <ul style="list-style-type: none"><li>• kein Label “NonRecipe” vorhanden</li><li>• Fusion ist eine gemischte Kategorie</li></ul> |

Fazit:

- Klassifikation ist möglich dem LLM
- Braucht vermutlich aber deutlich mehr Training für eine höhere Genauigkeit



# Instruction Fine Tuning

Datasets: AdamCodd/**recipe-nlg-alpaca**

like 0

Modalities: Text

Formats: json

Size: 100K - 1M

Libraries: Datasets

pandas

Croissant

+ 1

License:

cc-by-nc-4.0

Dataset card

Data Studio

Files and versions

xet

### You need to agree to share your contact information to access this dataset

This repository is publicly accessible, but you have to accept the conditions to access its files and content.

Your request to access this repository has been submitted and is awaiting a review from the repository authors. You can check the status of all your access requests in [your settings](#).

→ Copy to bucket **NEW**

Downloads last month

Total file size:  
150 MB

A heavily curated dataset from recipe-nlg (source="Gathered" only). A lot of scraping artifacts, typographical errors, unicode, empty and very short recipes were removed.

Then it has been formatted into Alpaca instruction set with Instructions, Input and Output. The total number of recipes went from ~2M2 (original dataset) to ~500K. Obviously, it's still not perfect (I won't lie, the original dataset was very flawed). To fully fix this would require a very time-consuming manual edition, so you can consider it a WIP.

If you want to support me, you can [here](#).

# EID Recipes - Eigener Instruction-Datensatz

Basierend auf RecipeNLG

- Aufgaben-Typen:
  1. Title\_To\_Recipe
    - nur der Titel ist gegeben
  2. Ingridients\_To\_Recipe
    - nur (Teil-)Zutaten
  3. Title\_Ingridients\_To\_Recipe
    - Titel + (Teil-)Zutaten

## Outputformat

**Recipe:** { Title }

**Ingridients:**

{ Ingridients[0] }

{ Ingridients[1] }

...

{ Ingridients[n] }

**Directions:**

{ Directions[0] }

{ Directions[1] }

...

{ Directions[n] }

# EID Recipes - Eigener Instruction-Datensatz

Basierend auf RecipeNLG

- Zutaten ohne Mengen
  - Modell soll sinnvolle Mengen lernen
- Zufälliger Zutaten-Drop
  - Modell soll fehlende Zutaten sinnvoll ergänzen
- Zutatenreihenfolge mischen
  - Modell soll die Zutatenliste nicht einfach in den Output durchreichen

## Phi-3-Format

**<|user|>**

All I've got is shrimp, mayo, onion, cream cheese, lemon juice, celery. What can I cook?

**<|assistant|>**

**Recipe:** Country Cafe'S Shrimp Spread

**Ingredients:**

8 oz. cream cheese, softened

1/2 c. mayo

1/4 c. chopped onion

1 rib chopped celery

1 tsp. lemon juice

1 small can shrimp, drained

salt and pepper to taste

**Directions:** Mix all of the ingredients with fork. Chill. Serve with crackers.

# EID Recipes - Aufgaben-Templates

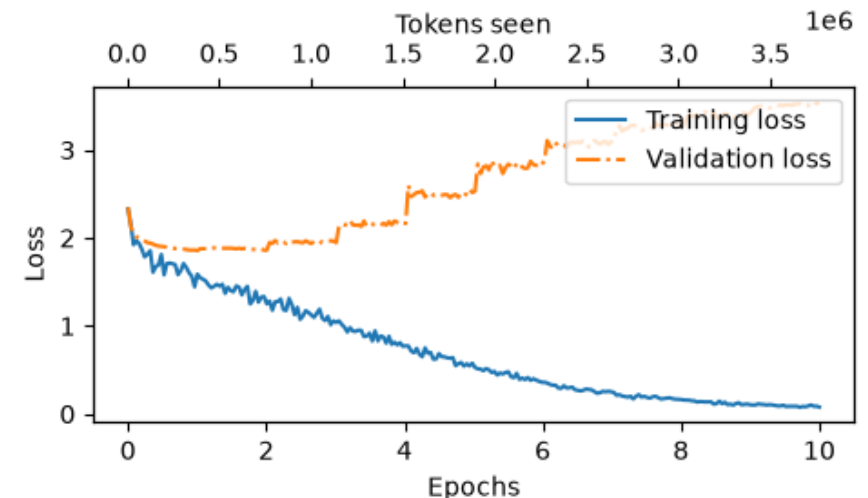
| Title_To_Recipe                                                                                                                                                                                     | Ingridients_To_Recipe                                                                                                                                                                                   | Title_Ingridiens_To_Recipe                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• How do I make {title}?</li><li>• I want to cook {title} tonight can you give me the full recipe?</li><li>• Whats the easiest way to make {title}?</li></ul> | <ul style="list-style-type: none"><li>• All I've got is {ingredients}. What can I cook?</li><li>• Make a recipe using {ingredients}.</li><li>• Suggest a dish I can make with {ingredients}?"</li></ul> | <ul style="list-style-type: none"><li>• I've got {ingredients} and I'm thinking {title}. Full recipe?</li><li>• create a recipe for {title} with {ingredients}.</li><li>• What's the recipe for {title} if I have {ingredients}?"</li></ul> |

# EID Recipes - Baseline Test & Training

| Test             | Input                                                                                                                            | Output                                                                                                                                | Kommentar                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Text Generierung | Make a recipe using salt, bread machine yeast, onion powder, onion, butter, brown sugar, bread flour, milk, black pepper, water. | If you want to add more salt, add more anise.<br>If you want to add more pepper, add more chili pepper.<br>If you want to add more... | ✓ Struktur okay<br>– Logik macht keinen Sinn |

- Train: 1500 | Val: 100 | Test: 100 Rezepte
- 10 Epochen

|            | Start Wert | End Wert |
|------------|------------|----------|
| Train      | 2,329      | 0,077    |
| Validation | 2,347      | 3,532    |



# EID Recipe - Evaluation

| Input                                                                                                                            | Pre-Training Response                                                                                                                 | Fine Tuned                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Kommentar                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Make a recipe using salt, bread machine yeast, onion powder, onion, butter, brown sugar, bread flour, milk, black pepper, water. | If you want to add more salt, add more anise.<br>If you want to add more pepper, add more chili pepper.<br>If you want to add more... | <b>Recipe:</b> Zucchini Bread<br><b>Ingredients:</b><br>1 lb. fresh zucchini<br>1/2 lb. dried zucchini<br>1 c. brown sugar<br>1/2 c. milk<br>1 tsp. salt<br>1 tsp. onion powder<br>1 tsp. dried bread crumbs<br>1 c. melted butter<br>1 c. chopped onion<br><b>Directions:</b><br>Cook zucchini in boiling water until tender.<br>Drain.<br>Melt butter in large skillet over medium heat?<br>Add zucchini; cook until tender.<br>Add sugar, brown sugar, milk, salt and onion powder.<br>Cook until thick.<br>Stir in bread crumbs and melted butter.<br>Add liquid from skillet and milk.<br>Cook until thick.<br>Can be frozen.... | <ul style="list-style-type: none"> <li>✓ Erkennt, dass es sich um ein Brot Rezept handelt</li> <li>✓ Saubere Struktur</li> <li>✓ Füllt mit passenden Zutaten auf</li> <li>- Vergisst Zutaten</li> <li>- Mengenangaben unlogisch</li> <li>- Anleitung unlogisch</li> </ul> |

# EID Recipe - LLM-as-a-Judge

| Modell Antwort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Judges (Llama-3.3-70B) Antwort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Recipe: Zucchini Bread</p> <p>Ingredients:</p> <ul style="list-style-type: none"><li>1 lb. fresh zucchini</li><li>1/2 lb. dried zucchini</li><li>1 c. brown sugar</li><li>1/2 c. milk</li><li>1 tsp. salt</li><li>1 tsp. onion powder</li><li>1 tsp. dried bread crumbs</li><li>1 c. melted butter</li><li>1 c. chopped onion</li></ul> <p>Directions:</p> <p>Cook zucchini in boiling water until tender.</p> <p>Drain.</p> <p>Melt butter in large skillet over medium heat?</p> <p>Add zucchini; cook until tender.</p> <p>Add sugar, brown sugar, milk, salt and onion powder.</p> <p>Cook until thick.</p> <p>Stir in bread crumbs and melted butter.</p> <p>Add liquid from skillet and milk.</p> <p>Cook until thick.</p> <p>Can be frozen....</p> | <p>Score von Llama 3:</p> <p>&gt;&gt; I would score this model response a 0.</p> <p>The response does not meet the requirements of the input in several ways:</p> <ol style="list-style-type: none"><li><b>**Missing ingredients**:</b> The recipe includes ingredients that were not mentioned in the input, such as zucchini, dried zucchini, and dried bread crumbs.</li><li><b>**Unused ingredients**:</b> The input mentions bread machine yeast, bread flour, black pepper, and water, which are not used in the provided recipe.</li><li><b>**Incorrect recipe**:</b> The recipe provided is for a zucchini bread, which does not match the ingredients listed in the input.</li><li><b>**Lack of logic**:</b> The recipe includes steps that are not logical, such as cooking zucchini in boiling water and then melting butter in a skillet to cook the zucchini again.</li><li><b>**Incompleteness**:</b> The recipe does not provide a complete or coherent set of instructions for making a bread using the ingredients listed in the input.</li></ol> <p>Overall, the response does not demonstrate an understanding of the input or the ability to generate a recipe that uses the provided ingredients in a logical and correct way.</p> |



# EID - LLM-as-a-Judge

- Richter Modell
  - **Llama-3.3-70B**
  - Sambanova-API
- Prompt
  - “Given the input {input}, score the model response {response} on a scale from 0 to 100, where 100 is the best score. Respond with the integer number only.”
- Bewertung
  - Ø **21,41** auf 100 Test Beispiel

# Instruction Fine Tuning

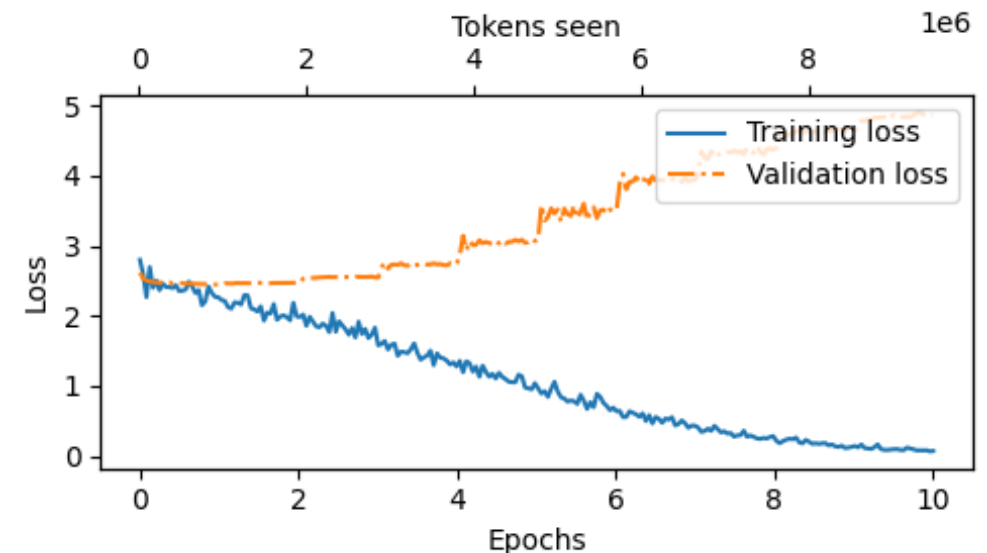
## LIMA

# LIMA - Baseline Test & Training

| Test             | Input                                                                                               | Output                                                                                  | Kommentar                             |
|------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------|
| Text Generierung | < user ><br>I am beginner of LaTeX.<br>From many examples I<br>found, I notice that it's<br>very... | I am beginner of LaTeX.<br>From many examples I<br>found, I notice that it's<br>very... | – Einfache Wiederholung<br>des Inputs |

- Train: 978 | Val: 52 | Test: 300
- 10 Epochen

|            | Start Wert | End Wert |
|------------|------------|----------|
| Train      | 2,797      | 0,069    |
| Validation | 2,615      | 4,868    |



# LIMA - LLM-as-a-Judge

- Richter Modell: **Llama-3.3-70B**
- Prompt: **“Given the input {input}, score the model response {response} on a scale from 0 to 100, where 100 is the best score. Respond with the integer number only.”**

# LIMA – Evaluation (LLM-as-a-Judge)

| Modell Antwort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Judges Antwort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>There are many eminent scholars of the Quechua language, which is spoken by the Chaco people of South America. Among them are:</p> <ul style="list-style-type: none"><li>* Hector Avalos-Navarro, Ph.D., Emeritus Professor of Linguistics, New York University</li><li>* Luís Figo da Silva, D.Phil., Visiting Professor of Linguistics, Madrid Institute for Advanced Studies</li><li>* Alfredo Pessoa, Ph.D., Emeritus Professor of Linguistics, Universidade de São Paulo</li><li>* <b>Alfredo V. Rodríguez-Galvão</b>, Ph.D., Professor of Linguistics, Universidade de São Paulo</li><li>* <b>Alfredo V. Rodríguez-Galvão</b>, D.Phil., Visiting Professor of Linguistics, Madrid Institute for Advanced Studies</li><li>* <b>Alfredo V. Rodríguez-Galvão</b>, Profesora de Linguagem, AT&amp;T Bell Laboratories</li><li>* <b>Alfredo V. Rodríguez-Galvão</b>, Profesora de Linguagem, Universidade de São Paulo</li><li>* Alfredo V. Vasconcelos, Ph.D., Professor of Linguistics, Madrid</li></ul> | <p>Score von Llama 3:<br/>&gt;&gt; <b>I would score this response a 0.</b></p> <p>Here's why:</p> <ol style="list-style-type: none"><li><b>**Incorrect information**</b>: The Quechua language is not spoken by the Chaco people. Quechua is an indigenous language family spoken in the Andean region of South America, primarily in Peru, Bolivia, Ecuador, and Chile. The Chaco people are an indigenous group living in the Chaco region of South America, but they do not speak Quechua.</li><li><b>**Non-existent scholars**</b>: A quick search reveals that none of the scholars mentioned in the response are real or notable experts in the field of Quechua language studies. The names, titles, and institutions appear to be fabricated.</li><li><b>**Lack of logic and coherence**</b>: The response lists multiple instances of the same person (Alfredo V. Rodríguez-Galvão) with different titles and affiliations, which is illogical and suggests a lack of effort in providing accurate information.</li><li><b>**Incompleteness**</b>: The response does not provide any accurate or relevant information about eminent scholars of the Quechua language.</li></ol> <p>Overall, the response demonstrates a complete lack of knowledge and understanding of the topic, and its inaccuracies and fabrications make it unusable. Therefore, I score it 0.</p> |

# LIMA - LLM-as-a-Judge

- Richter Modell
  - **Llama-3.3-70B**
- Prompt
  - “Given the input {input}, score the model response {response} on a scale from 0 to 100, where 100 is the best score. Respond with the integer number only.”
- Bewertung
  - Ø **11,01** auf 300 Test Beispiel

# LIMA - Datensatz

Mixed-Domain im Phi-3-Format

- **LIMA** → Less Is More for Alignment
- **1.030 Train / 300 Test**  
bewusst klein & hochwertig kuratiert

## LIMA(Phi-3)

**<|user|>**

What is currently considered normal that future generations will consider disgusting or immoral?

**<|assistant|>**

One current practice that future generations might consider disgusting or immoral is the contemporary form of meat consumption.

In the not-too-distant future, technological advances could enable synthetic meat production that will be indistinguishable from farm-grown meat in both taste and nutritional value. This method of meat production won't involve the suffering or killing of animals, and will have much less negative external effects (e.g. farmland areas, carbon emission). So, after this method becomes affordable to the masses, consuming meat that originated from livestock might be severely frowned upon, as there are equivalent alternatives with much less negative by-products.

# LIMA - Evaluation

| Input                                                  | Models Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Kommentar                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Who are some eminent scholars of the Quechua language? | <p>There are many eminent scholars of the Quechua language, which is spoken by the Chaco people of South America. Among them are:</p> <ul style="list-style-type: none"><li>* Hector Avalos-Navarro, Ph.D., Emeritus Professor of Linguistics, New York University</li><li>* Luís Figo da Silva, D.Phil., Visiting Professor of Linguistics, Madrid Institute for Advanced Studies</li><li>* Alfredo Pessoa, Ph.D., Emeritus Professor of Linguistics, Universidade de São Paulo</li><li>* <b>Alfredo V. Rodríguez-Galvão</b>, Ph.D., Professor of Linguistics, Universidade de São Paulo</li><li>* <b>Alfredo V. Rodríguez-Galvão</b>, D.Phil., Visiting Professor of Linguistics, Madrid Institute for Advanced Studies</li><li>* <b>Alfredo V. Rodríguez-Galvão</b>, Profesora de Linguagem, AT&amp;T Bell Laboratories</li><li>* <b>Alfredo V. Rodríguez-Galvão</b>, Profesora de Linguagem, Universidade de São Paulo</li><li>* Alfredo V. Vasconcelos, Ph.D., Professor of Linguistics, Madrid</li></ul> | <ul style="list-style-type: none"><li>✓ Auf dem ersten Blick gutes Ergebnisse</li><li>✓ Ausgabe sieht strukturell schön aus</li><li>✓ Sprachliche Zusammenhänge sind verständlich</li><li>– Selbe Person in unterschiedlichen Positionen</li><li>– Halluziniert faktisch und vom Inhalt her</li></ul> |



**RAG**

# RAG-Setup

- **Embedding:** bge-base-en-v1.5 (lokal via Ollama)
- **LLM:** Llama-3.2-1B-Instruct
- **Datensatz:** RecipeNLG

| Retrieval-Ergebnisse                                  |                                                                                                                                                                       |        |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Anfrage                                               | Top-Treffer                                                                                                                                                           | CS     |
| How to make a <b>chocolate cake</b> ?                 | <b>Recepie:</b> Favorite Chocolate Cake<br><b>Ingredients:</b> 1 3/4 c. flour, 2 eggs<br>...<br><b>Directions:</b> Sift all dry ingredients in a large mixing bowl... | 0.8346 |
| What are the ingredients for a <b>vegan lasagna</b> ? | <b>Recepie:</b> Sour Cream Lasagna<br><b>Ingredients:</b> 1 (8 oz.) pkg. egg dumpling noodles, 1 1/2 lb. ground beef, 1...<br><b>Directions:</b> Cook noodles...      | 0.7287 |
| How many Teams are in the <b>Bundesliga</b> ?         | <b>Recepie:</b> Beer Bread<br><b>Ingredients:</b> 3 c. self rising flour, 1 - 12 oz. can beer...<br><b>Directions:</b> ...                                            | 0.3965 |

CS = Kosinus Ähnlichkeit

# Modell mit vs. ohne RAG

*How do I make No-Bake Nut Cookies?*

- **Ohne RAG:**
  - halluziniert ein plausibel klingendes falsches Rezept
  - No-Bake Nut Cookies werden zu Haferflocken-Erdnussbutter-Honig-Kugeln



# Modell mit vs. ohne RAG

*How do I make No-Bake Nut Cookies?*

- **Mit RAG:**
  - Zutaten/Kochanweisung oft identisch mit dem äquivalenten Rezept
  - Eliminiert Halluzinationen nicht vollständig
- **Klare Dokumentenbasis**
- **Semi-Strukturierte Daten**
- **Semantische Ähnlichkeit**



Quelle: Nguyen 2024, Code a simple RAG from scratch; Gao et al. (2024), Retrieval-Augmented Generation for Large Language Models: A Survey

# How do I make No-Bake Nut Cookies?

| Eigenes Modell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | GPT-2 + Fine Tuning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Llama-3.2-1B-Instruct                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Llama + RAG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>For Diabetach</p> <p><b>Ingredients:</b></p> <p>3 eggs, lightly beaten<br/> 3 c. sugar<br/> 1 c. vegetable oil<br/> 4 tsp. all-purpose flour<br/> 1 tsp. vanilla extract<br/> 1 (6 oz.) pkg. (1 3/4 c.) semi-sweet chocolate morsels*<br/> 1 c. chopped walnuts (optional)</p> <p><b>Directions:</b></p> <p>In large bowl, combine the first 5 ingredients; combine the dry ingredients and next 4 ingredients. Pour batter into large balls (s removed) at a well-greased cookie sheet and press walnuts and bake at 350° for 10 - 15 minutes or until light and soft. Sprinkle powdered sugar on cookies after baking for 5 minutes.</p> <p>Do not overadd or any of the vanilla and continue baking for 5 minutes or until firm.</p> <p>Remove from oven; turn pierce cake immediately after 15 minutes. Cool completely on wire rack. Store at room for 6 weeks.</p> | <p><b>Recipe:</b> No-Bake Nut Cookies</p> <p><b>Ingredients:</b></p> <p>1 c. shortening<br/> 1 c. sugar<br/> 1 c. flaked coconut<br/> 1-1/2 c. brown sugar<br/> 3 eggs, slightly beat<br/> 1 (14 oz.) pkg. almonds, coarsely chopped<br/> 1 c. soy lecithin<br/> 4 c. sifted all-purpose flour<br/> 2 tsp. soda<br/> 1 tsp. salt<br/> 2 tsp. pet food seasoning<br/> 3 c. unsalted macadamia nuts, finely chopped</p> <p><b>Directions:</b></p> <p>Stir shortening; beat until smooth. Add sugar, eggs, nuts and soy lecithin. Blend until smooth. Drop tablespoonfuls (3 oz.) onto ungreased cookie sheets. Bake at 375° for 12 minutes. Let cool on wire rack.</p> | <p><b>**Ingredients:**</b></p> <p>* 2 cups rolled oats<br/> * 1 cup creamy peanut butter (or any other nut butter)<br/> * 1/2 cup confectioners' sugar<br/> * 1/4 cup honey or maple syrup (adjust to taste)<br/> * 1 teaspoon vanilla extract<br/> * 1/2 cup chopped nuts (walnuts, pecans, or hazelnuts work well)<br/> * Optional: chocolate chips or chopped dried fruit</p> <p><b>**Instructions:**</b></p> <ol style="list-style-type: none"> <li><b>**Prepare the ingredients**:</b> In a large mixing bowl, combine the oats and peanut butter. Mix until smooth.</li> <li><b>**Add sweetener and vanilla**:</b> Gradually add the confectioners' sugar and honey/maple syrup to the mixture. Stir until combined.</li> <li><b>**Add nuts and optional mix-ins**:</b> Fold in the chopped nuts and any desired mix-ins (chocolate chips or dried fruit).</li> <li><b>**Chill the dough**:</b> Cover the bowl with plastic wrap and refrigerate for at least 30 minutes to allow the flavors to meld.</li> <li><b>**Shape the cookies**:</b> Scoop small balls of dough onto a baking sheet lined with parchment paper, about 1 inch apart. You should end up with around 20-25 cookies.</li> <li><b>**Let them set**:</b> Refrigerate or freeze the cookie balls for at least 2 hours to firm up before serving.</li> </ol> | <p><b>**Ingredients:**</b></p> <p>* 2 cups of rolled oats<br/> * 1 cup of creamy peanut butter (or sunflower seed butter for a nut-free option)<br/> * 1/2 cup of honey or maple syrup (optional)<br/> * 1/4 cup of chopped walnuts, pecans, or any other nut you prefer<br/> * 1/4 teaspoon of salt<br/> * Optional: chocolate chips, dried cranberries, or other mix-ins of your choice</p> <p><b>**Instructions:**</b></p> <ol style="list-style-type: none"> <li>In a large mixing bowl, combine the oats and salt.</li> <li>Add in the peanut butter and honey (if using) and mix until well combined.</li> <li>If using nuts, fold them into the mixture at this stage.</li> <li>Cover the bowl with plastic wrap and refrigerate for at least 30 minutes to allow the flavors to meld together.</li> <li>Once chilled, use a cookie scoop or spoon to drop rounded tablespoon-sized balls of dough onto a baking sheet lined with parchment paper.</li> <li>If desired, press a chocolate chip or dried cranberry into each ball before ..</li> </ol> |

# Warum funktioniert das Rezept-RAG?

- Externe Wissensbasis
  - muss nicht antrainiert werden, sondern wird bei Bedarf abgerufen
- Klare Dokumenteinheiten
  - Jedes Rezept bleibt vollständig erhalten, Zutaten und Schritte werden nicht getrennt
- Semistrukturierte Daten
  - Recipe → Ingredients → Directions macht Inhalte eindeutig interpretierbar und ist nah am Outputformat
- Semantische Ähnlichkeit
  - Ähnliche Zutaten, Gerichtstypen und Kochmethoden führen zu ähnlichen Embeddings → Rezepte können gut verglichen werden



# Quellen

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# Beitrag zur Präsentation

| Teammitglied       | Folien                   | Thema                   |
|--------------------|--------------------------|-------------------------|
| Fabian Holländer   | 1-7; 17-20; 25-29; 41-48 | Datensätze; RAG         |
| Phillip Graph      | 30-40                    | Instruction Fine Tuning |
| Konstantin Schmidt | 7-16                     | Pre-Training            |